

AMENDMENT NO. 1 DECEMBER 2025
TO
IS 18158 : 2023 TEXTILE FLOOR COVERING — ARTIFICIAL GRASS CARPET
MADE OF SYNTHETIC YARN FOR LANDSCAPE — SPECIFICATION

[Page 2, Table 1, Sl No. (v), col (3)] — Substitute ‘500’ for ‘1 524’.

(Page 5, Annex B) — Substitute the following for the existing:

‘ANNEX B

[Table 1, Sl No. (v)]

TEST METHOD FOR WATER INFILTRATION RATE

B-1 PRINCIPLE

Water is ponded in a sealed cylindrical ring containing the test specimen. The infiltration rate is determined by measuring the fall of water level over time.

B-2 APPARATUS

B-2.1 Single Ring Infiltrometer — having internal diameter $300 \text{ mm} \pm 2 \text{ mm}$ and a method of sealing the ring to the product to be tested (either mechanically with a clamp or by use of a sealant) [see Fig. 2(A)].



FIG. 2(A) SINGLE RING INFILTROMETER



FIG. 2(B) SUPPORTING GRID

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FIG. 2(C) MOUNTING OF THE SPECIMEN ON SUPPORTING GRID

FIG. 2 SINGLE RING INFILTROMETER WITH SUPPORTING GRID

B-2.2 Support Grid — for underpinning the product to prevent it distorting when water is poured into the apparatus [see Fig. 2(B) and Fig. 2(C)]. The maximum deformation of the product shall be 5 mm from the outside of the ring to the centre once water has been added to the ring. Within the ring there shall be three parallel horizontal wires or bars and a central bar perpendicular to the other three [see Fig. 2(B)]. These shall be $2.5 \text{ mm} \pm 0.5 \text{ mm}$ in width to prevent obstruction of any drainage holes in the carpet surface.

B-2.3 Stopwatch — with a resolution of 0.1 s for recording the time taken for the water level to fall between the specified markers.

B-2.4 Spirit Level — to check that the infiltration ring is placed horizontally before starting the test and to ensure the uniform water distribution across the specimen.

B-2.5 Graduated Scale — readable to an accuracy of 1 mm, to measure the depth of water.

B-2.6 Thermometer — capable of measuring to $\pm 1 \text{ }^{\circ}\text{C}$

B-2.7 Geotextile or Mesh Grid — with maximum aperture size $300 \text{ }\mu\text{m}$

B-3 TEST SPECIMEN

Select and prepare a test specimen of the carpet surface ensuring that the minimum number of any drainage holes are visible within the 300 mm ring area. Measure the diameter and location of the holes within the test piece and record and/or photograph the evidence, which may be included in the report.

B-4 CONDITIONING

Condition the test samples and any relevant infill in the laboratory at a temperature of $27 \text{ }^{\circ}\text{C} \pm 2 \text{ }^{\circ}\text{C}$ at a relative humidity of (65 ± 2) percent for a minimum period of 4 h.

B-5 PROCEDURE

B-5.1 Ensure that the infiltration ring is placed in a perfectly horizontal position using a spirit level before starting the test, so that the water head is uniform across the specimen.

B-5.2 Measure and record the temperature of the test water with a thermometer accurate to $\pm 1 \text{ }^{\circ}\text{C}$. This value shall be included in the test report.

B-5.3 Apply water evenly onto the surface of the specimen through a geotextile or mesh grid. The water head shall be maintained between 70 mm and 90 mm above the top of the specimen.

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B-5.4 Mark the inside wall of the infiltration ring at positions $30 \text{ mm} \pm 1 \text{ mm}$ and $10 \text{ mm} \pm 1 \text{ mm}$ above the top of the specimen. These marks define the start and finish levels for timing the infiltration.

B-5.5 Measure the time required for the water level to fall from the 30 mm mark to the 10 mm mark, recording the value to the nearest 0.1 s. The geotextile or mesh disc shall be removed before starting the timer. If the infiltration is very slow, the test shall be stopped after 30 min even if the 10 mm level has not been reached.

B-5.6 Repeat the test a further two times under the same conditions, giving a total of three test results.

B-6 CALCULATION

Calculate the water infiltration rate I_A , in mm/h, from the following formula:

$$I_A = \frac{F_{WA}}{t_A}$$

where

I_A = water infiltration rate;

F_{WA} = fall of water level (mm); and

t_A = time taken for the water level to fall in hour (h).

B-8 TEST REPORT

The report shall include:

- a) reference to this standard;
- b) complete identification and photograph(s) of the surface tested;
- c) water temperature;
- d) any deviations from the procedure;
- e) any unusual features observed;
- f) mean average of the test results;
- g) individual test result(s), including a reference to the clause which explains how the results were calculated; and
- h) date of the test.'